

ANALYSIS OF A GEOMETRICAL MULTISCALE MODEL BASED ON THE COUPLING OF ODEs AND PDEs FOR BLOOD FLOW SIMULATIONS*

ALFIO QUARTERONI[†] AND ALESSANDRO VENEZIANI[‡]

Abstract. In hemodynamics, local phenomena, such as the perturbation of flow pattern in a specific vascular region, are strictly related to the global features of the whole circulation (see, e.g., [L. Formaggia et al., *Comput. Vis. Sci.*, 2 (1999), pp. 75–83]). In [A. Quarteroni, S. Ragni, and A. Veneziani, *Comput. Vis. Sci.*, 4 (2001), pp. 111–124] we have proposed a heterogeneous model, where a local, accurate, three-dimensional description of blood flow by means of the Navier–Stokes equations in a specific artery is coupled with a systemic, zero-dimensional, lumped model of the remainder of circulation. This is a geometrical multiscale strategy, which couples an initial-boundary value problem to be used in a specific vascular region with an initial-value problem in the rest of the circulatory system. It has been successfully adopted to predict the outcome of a surgical operation (see [K. Laganà et al., *Biorheology*, 39 (2002), pp. 359–364, G. Dubini et al., *Proceedings of the European Congress on Computational Methods in Applied Sciences and Engineering*, Barcelona, Spain, 2000]). However, its interest goes beyond the context of blood flow simulations. In this paper we provide a well-posedness analysis of this multiscale model by proving a local-in-time existence result based on a fixed-point technique. Moreover, we investigate the role of matching conditions between the two submodels for the numerical simulation.

Key words. geometrical multiscale models, blood flow simulations, fixed-point techniques

AMS subject classifications. 35M20, 35Q30, 76D03, 65L05, 76Z05

PII. S1540345902408482

1. Introduction. In computational fluid dynamics there might be the need for accurately simulating the flow in a subregion of a complex system that can be reasonably represented by a hydraulic network. One may think, for instance, of a system for water supply for which there is the interest of ascertaining the spatial variations of velocity and pressure only in a specific pipeline or in a reservoir. Another instance, which has motivated our investigation, is the blood circulatory system. Here the underlying (low cost) model is based on an analogy with electric circuits and can predict the time evolution of average physical quantities (flow rate and pressure) in the different compartments of the overall system (see, e.g., [15, 27]). Yet, there is the interest of carrying out three-dimensional (3D) simulations on a sensible region (e.g., a coronary bypass, a carotid bifurcation, a stented artery, etc.). In these cases, the local investigation requires the specification of boundary conditions on the interface between the region of interest and the remainder of the network. In fact, the significance of numerical results is strictly related to the capability of the numerical device of properly accounting for the exchange of information with the global network. Actually, it is well known that, in the circulatory system, local behavior of blood flow in a specific region is strictly related to the systemic features of the circulation. For

*Received by the editors May 28, 2002; accepted for publication (in revised form) December 18, 2002; published electronically May 1, 2003. This work was supported by Cofin2000 MURST Project, by the Special Project “Large Scale Computing: Multiscale Computing in Biofluidynamics” of the Politecnico di Milano, and by Swiss National Science Foundation Project 21-54139.98.

<http://www.siam.org/journals/mms/1-2/40848.html>

[†]IMA and Institute Bernoulli, École Polytechnique Fédérale de Lausanne, CH-1024 Switzerland (alfio.quarteroni@epfl.ch).

[‡]MOX (Modeling and Scientific Computing), Dipartimento di Matematica “F. Brioschi,” Politecnico di Milano, Via Bonardi 9, I - 20133 Milano, Italy (veneziani@mate.polimi.it).

instance, the presence of a plaque localized in a carotid bifurcation can induce a flow rate increment in other vessels in order to restore sufficient oxygenation of the brain, thus modifying the local flow features (see [1]).

Blood is a complex fluid, which can be described by means of the incompressible Navier–Stokes equations, possibly accounting for a non-Newtonian rheology (see, e.g., [17, 29]). Numerical simulations of the whole circulatory system completely based on the numerical solution of the Navier–Stokes equations are unaffordable. Actually, a tremendous amount of morphological data would be required, which typically are not available, not to mention the computational costs of such investigations, which would be prohibitive. In order to set up numerical tools for the correct simulation of blood flow, in [7] we have proposed a *geometrical multiscale* approach. In fact, a hierarchy of models was introduced with a different level of accuracy, ranging from the local 3D accurate Navier–Stokes equations to systems of ODEs: the latter are associated with lumped parameter models set up by exploiting the analogy between hydraulic and electric networks (see, e.g., [27]) and account for the systemic circulation. These different models are then coupled at the numerical level in order to provide at the same time local details and global information on blood flow. The same strategy has been successfully used in [5] for the coupling of a 3D model with a one-dimensional model based on a nonlinear hyperbolic system that accounts for the propagation of pressure waves in the arterial tree.

A heterogeneous model which couples ordinary differential systems for the whole circulation with the 3D Navier–Stokes equations has been proposed in [16] and successfully adopted in a real-life application arising from pediatric surgery (see [2, 3]).

The aim of the present work is to provide a theoretical analysis of this heterogeneous model coupling ODEs and PDEs, providing a local-in-time existence result for the solution. The heterogeneous problem will actually be split into subproblems (the system of ODEs for the global description and the PDE system based on the Navier–Stokes equations for the local model). The solution of the original problem will be regarded as the solution of a suitable fixed point problem, based on the successive solution of the subproblems. A priori estimates for the solutions of the individual subproblems will be stated at first. Then we will prove that the classical Schauder fixed-point theorem can be applied in order to prove the existence of the solution of the original problem.

The outline of this paper is as follows. In section 2 we address the analysis of the Navier–Stokes problem, having provided boundary conditions which can be considered as a generalization of the mean pressure drop problem investigated in [10], as they arise in bioengineering applications. The well-posedness analysis carried out for the mean pressure drop problem is therefore generalized to the present case. In section 3 we discuss the basic features of the ordinary differential systems arising in the systemic modeling. We will address the lumped parameter description of a simple compliant pipe and then extend the basic concepts to the case of a nonlinear ODE system accounting for the whole circulation. In section 4 we deal with the heterogeneous model obtained by coupling the local and the systemic submodels. In section 5 we address the well-posedness analysis and prove our main result. We comment on the way the interface conditions can be split to set up an iterative algorithm for the numerical solution of the coupled multiscale problem. Finally, in section 6 we provide some numerical results on academic test cases as well as on a case of real-life clinical interest.

1.1. Basic notations. Let Ω be a bounded domain in $\mathbb{R}^d$ ($d = 2, 3$). We denote by $L^2(\Omega)$ the space of real functions whose square is integrable in Ω , and by $(\cdot, \cdot)$

and $\|\cdot\|$ the associated inner product and norm, respectively. The corresponding vector space $(L^2(\Omega))^d$ will be denoted by $\mathbf{L}^2(\Omega)$. Similarly, we introduce the space $H^1(\Omega) = \{v \in L^2(\Omega) | \nabla v \in \mathbf{L}^2(\Omega)\}$ whose norm is denoted by $\|\cdot\|_1$. Correspondingly, we set $\mathbf{H}^1(\Omega) = (H^1(\Omega))^d$, and we still denote its norm by $\|\cdot\|_1$. For functions depending on space and time, for a given space $\mathbf{V}$ of space dependent functions, we define (for some $T > 0$)

$$L^2(0, T; \mathbf{V}) = \left\{ \mathbf{v} : (0, T) \rightarrow \mathbf{V} \mid \mathbf{v} \text{ is measurable and } \int_0^T \|\mathbf{v}(t)\|_{\mathbf{V}}^2 dt < \infty \right\}$$

with norm $\|\mathbf{v}\|_{L^2(0, T; \mathbf{V})} = (\int_0^T \|\mathbf{v}(t)\|_{\mathbf{V}}^2 dt)^{1/2}$ and

$$L^\infty(0, T; \mathbf{V}) = \left\{ \mathbf{v} : (0, T) \rightarrow \mathbf{V} \mid \sup_{t \in (0, T)} \|\mathbf{v}(t)\|_{\mathbf{V}} < \infty \right\}$$

with norm $\|\mathbf{v}\|_{L^\infty(0, T; \mathbf{V})} = \sup_{t \in (0, T)} \|\mathbf{v}(t)\|_{\mathbf{V}}$.

If Σ is an open nonempty subset of $\partial\Omega$, we denote by $H^{1/2}(\Sigma)$ the space of functions defined on Σ which are traces of functions in $H^1(\Omega)$. We recall that the trace operator $\mathcal{L}_{\text{trace}} : H^1(\Omega) \rightarrow H^{1/2}(\partial\Omega)$ is surjective and continuous (see [12]).

When considering functions which depend only on time, we define the space

$$\mathbf{L}^\infty(0, T) = \left\{ \mathbf{z} : (0, T) \rightarrow \mathbb{R}^m \mid \sup_{t \in (0, T)} |\mathbf{z}(t)| < \infty \right\}$$

endowed with the norm $\|\mathbf{z}\|_{\mathbf{L}^\infty(0, T)} = \sup_{t \in (0, T)} |\mathbf{z}(t)|$.

2. The local submodel. Let $\Omega \subset \mathbb{R}^3$ (or $\mathbb{R}^2$) be a vascular region where we aim at providing a detailed flow analysis. For each $\xi \in \Omega$, and at any time $t > 0$, we denote by $\mathbf{u}(\xi, t)$ and $p(\xi, t)$ the blood velocity and pressure, respectively. Moreover, we assume that blood is an incompressible Newtonian fluid. Correspondingly, we denote by ρ the blood density and by ν the blood viscosity, which are both assumed to be constant. The hypothesis of a Newtonian rheology is acceptable in large- and medium-size vessels, and it is supported by numerical evidence (see, e.g., [17]). To what extent the analysis that follows could be affected by a non-Newtonian rheology is, however, addressed in Remark 5.4.

Under the previous assumptions, blood flow can be described by the Navier–Stokes equations

$$(2.1) \quad \begin{cases} \rho \frac{\partial \mathbf{u}}{\partial t} + \rho(\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla \cdot \boldsymbol{\Sigma} = \mathbf{0}, \\ \nabla \cdot \mathbf{u} = 0, \end{cases} \quad \xi \in \Omega, t > 0,$$

where $\boldsymbol{\Sigma}$ is the Cauchy stress tensor, which in the case of a Newtonian fluid reads $\boldsymbol{\Sigma} = p\mathbf{I} - \nu(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$, $\mathbf{I}$ being the identity tensor. Equation (2.2) expresses momentum and mass conservation principles. We are neglecting the presence of any external force, which is reasonable in our context. If we plug the divergence-free constraint into the momentum equation, (2.1) reads

$$(2.2) \quad \begin{cases} \rho \frac{\partial \mathbf{u}}{\partial t} + \rho(\mathbf{u} \cdot \nabla) \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mathbf{0}, \\ \nabla \cdot \mathbf{u} = 0, \end{cases} \quad \xi \in \Omega, t > 0.$$

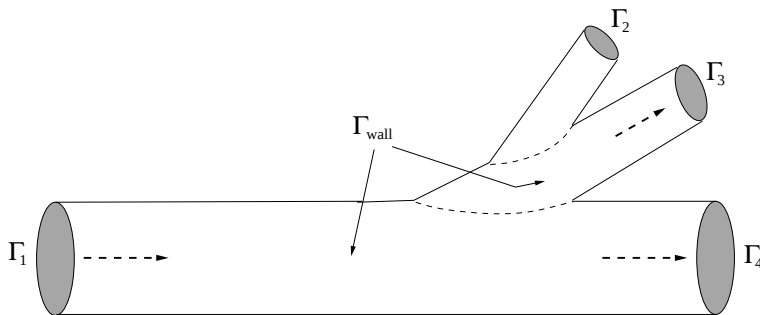


FIG. 2.1. A typical vascular region, where the boundary Γ_1 represents the upstream section, while Γ_i ($i = 2, 3, 4$) are the downstream sections.

For the sake of simplicity, we normalize ρ to 1. The previous system has to be provided with initial condition

$$(2.3) \quad \mathbf{u}(\boldsymbol{\xi}, 0) = \mathbf{u}_0(\boldsymbol{\xi}), \quad \boldsymbol{\xi} \in \Omega,$$

and boundary conditions. In this respect, we split $\partial\Omega$ into different parts (see Figure 2.1). The vascular wall will be denoted by Γ_{wall} . In the present work, we are assuming that the walls are rigid so that no-slip boundary condition

$$(2.4) \quad \mathbf{u}(\boldsymbol{\xi}, t) = 0, \quad \boldsymbol{\xi} \in \Gamma_{wall}, t > 0,$$

holds. The other parts of $\partial\Omega$ are the artificial boundaries which bound the computational domain. For the sake of clarity, we distinguish the “upstream” or proximal sections on the side of the heart, and the “downstream” or distal sections on the side of the peripheral vessels. We will suppose in general that there are $\bar{n}$ artificial sections, denoted by Γ_i ($i = 1, \dots, \bar{n}$), where we can prescribe different kind of boundary conditions corresponding to different specific boundary problems (see, e.g., [10, 5, 6, 26]). We could prescribe, in particular, the condition

$$(2.5) \quad \boldsymbol{\Sigma} \cdot \mathbf{n} - K_i(\mathbf{u} \cdot \mathbf{n})\mathbf{n} = p_i\mathbf{n} \quad \text{on } \Gamma_i,$$

where $p_i = p_i(t)$ is assumed to be a given function of t (independent of $\mathbf{x}$) on each section, $\mathbf{n}$ represents the outward unit vector on every part of the vessel boundary, and K_i is a suitable nonnegative constant. A different set of boundary conditions can be considered as well:

$$(2.6) \quad p\mathbf{n} - \nu \nabla \mathbf{u} \cdot \mathbf{n} - K_i(\mathbf{u} \cdot \mathbf{n})\mathbf{n} = p_i\mathbf{n} \quad \text{on } \Gamma_i.$$

When $K_i = 0$, (2.6) reduces to the usual Neumann or natural conditions associated with (2.2). In this case, in [10] it is pointed out that (2.6) should be preferred to (2.5) for numerical reasons. In what follows, we will refer therefore to (2.6). When $K_i = 0$ and p_i is independent of the space variables, we recover from conditions (2.6) the *mean pressure drop problem* investigated in [10], specifically in the context of hemodynamics in [26]. The physical justification for the case in which $K_i \neq 0$ is provided in what follows (see Figure 4.1).

Both the mathematical analysis and the numerical treatment of the Navier–Stokes problem are based on its weak formulation (see, e.g., [9]). We define the following

bilinear and trilinear forms:

$$(2.7a) \quad \begin{aligned} a : \mathbf{H}^1(\Omega) \times \mathbf{H}^1(\Omega) &\rightarrow \mathbb{R} \text{ such that (s.t.)} \\ a(\mathbf{v}, \varphi) &= \nu(\nabla \mathbf{v}, \nabla \varphi) + \sum_{i=1}^{\bar{n}} K_i \int_{\Gamma_i} (\mathbf{v} \cdot \mathbf{n}) \mathbf{n} \cdot \varphi d\gamma, \end{aligned}$$

$$(2.7b) \quad b : \mathbf{H}^1(\Omega) \times \mathbf{H}^1(\Omega) \times \mathbf{H}^1(\Omega) \rightarrow \mathbb{R} \text{ s.t. } b(\mathbf{v}, \mathbf{w}, \varphi) = ((\mathbf{v} \cdot \nabla) \mathbf{w}, \varphi),$$

$$(2.7c) \quad l : Q \times \mathbf{H}^1(\Omega) \rightarrow \mathbb{R} \text{ s.t. } l(\psi, \varphi) = -(\psi, \nabla \cdot \varphi),$$

where $Q = L^2(\Omega)$. Moreover, we set

$$(2.7d) \quad c(t, \varphi) = - \sum_{i=1}^{\bar{n}} p_i(t) \int_{\Gamma_i} \mathbf{n} \cdot \varphi d\gamma$$

for each $\varphi \in \mathbf{H}^1(\Omega)$. Furthermore, we define

$$(2.8) \quad \mathbf{V} = \{\varphi \in \mathbf{H}^1(\Omega) \mid \varphi|_{\Gamma_{wall}} = \mathbf{0}\}, \quad \mathbf{V}^* = \{\varphi \in \mathbf{V} \mid \nabla \cdot \varphi = 0\}.$$

The weak formulation of the problem given by (2.2), (2.3), (2.4), (2.6) reads as follows.

PROBLEM 2.1. *Given $\mathbf{u}_0 \in \mathbf{V}^*$ and $p_i \in L^\infty(0, T)$ for $i = 1, \dots, \bar{n}$, find $\mathbf{u} \in L^2(0, T; \mathbf{V})$ and $p \in L^2(0, T, Q)$ such that for all $t > 0$*

$$\begin{cases} \left(\frac{\partial \mathbf{u}}{\partial t}, \varphi \right) + a(\mathbf{u}, \varphi) + b(\mathbf{u}, \mathbf{u}, \varphi) + l(p, \varphi) = c(t, \varphi) & \forall \varphi \in \mathbf{V}, \\ l(\psi, \mathbf{u}) = 0 & \forall \psi \in Q \end{cases}$$

with $\mathbf{u}|_{t=0} = \mathbf{u}_0$. Its solenoidal version, which is more suitable for our analysis (see [10, 25]), reads as follows.

PROBLEM 2.2. *Given $\mathbf{u}_0 \in \mathbf{V}^*$, find $\mathbf{u} \in L^2(0, T; \mathbf{V}^*)$ such that for all $t \geq 0$*

$$(2.9) \quad \left(\frac{\partial \mathbf{u}}{\partial t}, \varphi \right) + a(\mathbf{u}, \varphi) + b(\mathbf{u}, \mathbf{u}, \varphi) = c(\varphi) \quad \forall \varphi \in \mathbf{V}^*$$

with $\mathbf{u}|_{t=0} = \mathbf{u}_0$.

2.1. Well-posedness analysis. Let $\mathbf{V}_1^*$ be the completion of $\mathbf{V}^*$ with respect to the norm of $\mathbf{L}^2(\Omega)$. For each $\mathbf{f} \in \mathbf{V}_1^*$ there exists exactly one $\mathbf{w} \in \mathbf{V}^*$ satisfying

$$(2.10) \quad (\nabla \mathbf{w}, \nabla \varphi) = (\mathbf{f}, \varphi) \quad \forall \varphi \in \mathbf{V}^*.$$

Conversely, for every $\mathbf{w} \in \mathbf{V}^*$ there exists at most one $\mathbf{f} \in \mathbf{V}_1^*$ such that (2.10) holds. In this way, as stated in [10, sect. 6], a one-to-one correspondence can be defined between the elements $\mathbf{f}$ of $\mathbf{V}_1^*$ and the functions $\mathbf{w}$ belonging to a suitable subspace of $\mathbf{V}^*$ that we denote by $D(\tilde{\Delta})$. The Stokes operator $\tilde{\Delta} : D(\tilde{\Delta}) \rightarrow \mathbf{V}_1^*$ is defined setting $-\tilde{\Delta} \mathbf{w} = \mathbf{f}$ so that (2.10) is satisfied. The inverse operator $\tilde{\Delta}^{-1}$ is self-adjoint and has a sequence of eigenfunctions $\{\mathbf{a}_k\}$, which are orthogonal in $\mathbf{V}^*$.

In what follows, c_i ($i = 1, 2, \dots$) will denote generic constants, not necessarily the same at different places. The inequalities

$$(2.11) \quad \sup_{\Omega} |\mathbf{w}| \leq c_1 \|\nabla \mathbf{w}\|^{1/2} \|\tilde{\Delta} \mathbf{w}\|^{1/2},$$

$$(2.12) \quad \|\nabla \mathbf{w}\| \leq c_2 \|\tilde{\Delta} \mathbf{w}\|$$

are satisfied for each $\mathbf{w} \in D(\tilde{\Delta})$, provided Ω is a bounded domain (see [10, 30]). Furthermore, if we define

$$\tilde{p} = \sup_{t \geq 0} \sum_{i=1}^{\bar{n}} |p_i|, \quad K = \max_i K_i,$$

the following estimates hold for every $\mathbf{w} \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_N\}$:

$$(2.13) \quad \left| \sum_{i=1}^{\bar{n}} p_i \int_{\Gamma_i} \tilde{\Delta} \mathbf{w} \cdot \mathbf{n} d\gamma \right| \leq c_3 \tilde{p} \|\tilde{\Delta} \mathbf{w}\|,$$

$$(2.14) \quad \left| \sum_{i=1}^{\bar{n}} K_i \int_{\Gamma_i} (\mathbf{w} \cdot \mathbf{n}) \mathbf{n} \cdot \tilde{\Delta} \mathbf{w} d\gamma \right| \leq c_4 K \|\nabla \mathbf{w}\| \|\tilde{\Delta} \mathbf{w}\|.$$

The former can be found in [10]. To prove (2.14) we start noting that for each $v \in H^1(\Omega)$ with null trace on Γ_{wall} the identity

$$(2.15) \quad \int_{\partial\Omega} v \mathbf{n} \cdot \tilde{\Delta} \mathbf{w} = \sum_{i=0}^{i=\bar{n}} \int_{\Gamma_i} v \mathbf{n} \cdot \tilde{\Delta} \mathbf{w} = \int_{\Omega} v \nabla \cdot \tilde{\Delta} \mathbf{w} + \int_{\Omega} \nabla v \cdot \tilde{\Delta} \mathbf{w} = \int_{\Omega} \nabla v \cdot \tilde{\Delta} \mathbf{w}$$

holds since $\tilde{\Delta} \mathbf{w}$ is a solenoidal function.

Indeed, let $v \in H^1(\Omega)$ satisfy

$$\begin{aligned} -\Delta v &= 0 \quad \text{in } \Omega, \\ v|_{\Gamma_i} &= K_i(\mathbf{w} \cdot \mathbf{n})|_{\Gamma_i}, \quad i = 1, \dots, \bar{n}, \\ v|_{\Gamma_{wall}} &= 0. \end{aligned}$$

v is actually the harmonic extension of the boundary data $K_i(\mathbf{w} \cdot \mathbf{n})|_{\Gamma_i}$. From well-known results on harmonic extensions we have (see, e.g., [18, Chap. 4, p. 104])

$$\|\nabla v\| \leq \sum_{i=1}^{\bar{n}} K_i \|\mathbf{w} \cdot \mathbf{n}\|_{H^{1/2}(\Gamma)} \leq c_4 K \|\nabla \mathbf{w}\|,$$

the latter inequality being a consequence of the Poincaré and the trace inequalities. Inequality (2.14) follows therefore from (2.15).

Now, we can prove the following proposition.

PROPOSITION 2.1. *Let ν and R be such that*

$$(2.16) \quad \kappa = \nu - 2Rc_2c_4 > 0.$$

Then there is a time interval on which the solution of Problem 2.2 does exist. Furthermore, if we assume that the initial and boundary data are sufficiently small, precisely

$$(2.17) \quad \|\nabla \mathbf{u}_0\| < \frac{\kappa}{4c_1\sqrt{c_2}} \quad \text{and} \quad \tilde{p} < \frac{\sqrt{\nu\kappa^3}}{\sqrt{32c_2^3c_1c_3}},$$

then the solution of Problem 2.2 exists for all $t > 0$ and satisfies the inequality

$$(2.18) \quad \|\nabla \mathbf{u}(t)\| \leq \frac{\kappa}{4c_1\sqrt{c_2}}.$$

Finally, there exists an interval $(0, T)$ in which the solution is unique and depends continuously on the data.

Proof. This proposition is an extension of a similar result holding for the mean pressure drop problem (Theorem 6 in [10]). Actually, as we have already pointed out, the mean pressure drop problem can be recovered as a particular case of Problem 2.2, when $K = 0$. We therefore report only the main steps of the proof.

Consider the sequence

$$\mathbf{u}_N = \sum_{k=1}^N \gamma_{kN}(t) \mathbf{a}_k$$

which is built by solving (2.9) for each $\varphi \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_N\}$. The solution $\mathbf{u}$ is obtained as the weak limit of a subsequence of $\{\mathbf{u}_N\}$ in $L^2(0, T; \mathbf{V}^*)$. The outline of the proof is the following: we first obtain some a priori estimates for $\{\mathbf{u}_N\}$ that are inherited by the limit solution when $N \rightarrow \infty$, yielding the existence of a solution locally in time. Then, if the initial data are small enough, we will prove further a priori estimates, ensuring the existence of the solution for all $t > 0$. Finally, we will prove that, locally in time, the limit solution is unique and depends continuously on the data.

Setting $\varphi = -\tilde{\Delta} \mathbf{u}_N$ in (2.9), we have

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|\nabla \mathbf{u}_N\|^2 + \nu \|\tilde{\Delta} \mathbf{u}_N\|^2 &= ((\mathbf{u}_N \cdot \nabla) \mathbf{u}_N, \tilde{\Delta} \mathbf{u}_N) + \sum_{i=1}^{\bar{n}} p_i \int_{\Gamma_i} \mathbf{n} \cdot \tilde{\Delta} \mathbf{u}_N \\ &\quad + \sum_{i=1}^{\bar{n}} K_i \int_{\Gamma_i} (\mathbf{u}_N \cdot \mathbf{n}) \mathbf{n} \cdot \tilde{\Delta} \mathbf{u}_N \\ &\leq \sup_{\Omega} |\mathbf{u}_N| \|\nabla \mathbf{u}_N\| \|\tilde{\Delta} \mathbf{u}_N\| + c_3 \tilde{p} \|\tilde{\Delta} \mathbf{u}_N\| \\ &\quad + c_4 K \|\nabla \mathbf{u}_N\| \|\tilde{\Delta} \mathbf{u}_N\| \\ &\leq c_1 \sqrt{c_2} \|\nabla \mathbf{u}_N\| \|\tilde{\Delta} \mathbf{u}_N\|^2 + \frac{c_3^2}{2\nu} \tilde{p}^2 + \frac{\nu}{2} \|\tilde{\Delta} \mathbf{u}_N\|^2 \\ &\quad + c_4 c_2 R \|\tilde{\Delta} \mathbf{u}_N\|^2. \end{aligned}$$

Therefore

$$(2.19) \quad \frac{d}{dt} \|\nabla \mathbf{u}_N\|^2 + (\kappa - 2c_1 \sqrt{c_2} \|\nabla \mathbf{u}_N\|) \|\tilde{\Delta} \mathbf{u}_N\|^2 \leq \frac{c_3^2}{\nu} \tilde{p}^2.$$

Using standard arguments (see [10, Thm. 6]), from this inequality we can infer that the Galerkin approximant $\mathbf{u}_N$ satisfies the following a priori estimate:

$$(2.20) \quad \|\mathbf{u}_N\|_{\mathbf{V}} \leq c\tilde{p}.$$

When $N \rightarrow \infty$, the same inequality is satisfied, locally in time, by the limit solution $\mathbf{u}$. This proves the local existence of a solution.

Furthermore, suppose that assumptions (2.16) and (2.17) hold. Then there exists a time interval on which the coefficient of $\|\tilde{\Delta} \mathbf{u}_N\|^2$ of (2.19) is positive. Exploiting (2.12), we can prove that in the same interval

$$\|\nabla \mathbf{u}_N\| < \frac{\kappa}{4c_1 \sqrt{c_2}}.$$

The same inequality holds for the limit solution obtained for $N \rightarrow \infty$. It is then possible to prove that the sup of the time interval on which (2.18) holds, i.e., the time instant at which (2.18) becomes an equality, is not finite, yielding the global existence result.

Consider now the issue of uniqueness and continuous dependence on the data. We first prove that there exists a time interval of continuous dependence on the data, assuming the uniqueness of the solution. Later, we will prove that the uniqueness actually holds. Suppose, therefore, that there exists a time interval where the solution of Problem 2.2 is unique. In particular, denote by $\mathbf{u}^{(1)}$ the solution associated with the data $p_i^{(1)}$ ($i = 1, \dots, \bar{n}$) and, correspondingly, by $\mathbf{u}^{(2)}$ the solution associated with $p_i^{(2)}$ ($i = 1, \dots, \bar{n}$). Set

$$\mathbf{w} = \mathbf{u}^{(1)} - \mathbf{u}^{(2)}, \quad \delta\tilde{p} = \max_i |p_i^{(1)} - p_i^{(2)}|.$$

By subtraction and some classical estimates (see [10]), we have

$$\begin{aligned} & \frac{1}{2} \frac{d}{dt} \|\mathbf{w}\|^2 + \nu \|\nabla \mathbf{w}\|^2 + \sum_i K_i \int_{\Gamma_i} |\mathbf{w} \cdot \mathbf{n}|^2 d\gamma \\ &= -b(\mathbf{u}^{(2)}, \mathbf{w}, \mathbf{w}) - b(\mathbf{w}, \mathbf{u}^{(2)}, \mathbf{w}) - b(\mathbf{w}, \mathbf{w}, \mathbf{w}) + \int_{\Gamma_i} (p_i^{(1)} - p_i^{(2)}) \mathbf{w} \cdot \mathbf{n} d\gamma \\ (2.21) \quad & \leq \frac{\nu}{2} \|\nabla \mathbf{w}\|^2 + \frac{c_6}{\nu} \sup |\mathbf{u}^{(2)}|^2 \|\mathbf{w}\|^2 + \frac{c_7}{\nu^3} \|\nabla \mathbf{u}^{(2)}\|^4 \|\mathbf{w}\|^2 + c_8 \|\mathbf{w}\| \|\nabla \mathbf{w}\|^2 + c_9 \delta\tilde{p}^2. \end{aligned}$$

Hence,

$$\begin{aligned} (2.22) \quad & \frac{1}{2} \frac{d}{dt} \|\mathbf{w}\|^2 + (\nu - c_{10} \|\mathbf{w}\|) \|\nabla \mathbf{w}\|^2 \\ & \leq c_{11} \left(\frac{1}{\nu} \sup |\mathbf{u}^{(2)}|^2 + \frac{1}{\nu^3} \|\nabla \mathbf{u}^{(2)}\|^4 \right) \|\mathbf{w}\|^2 + c_9 \delta\tilde{p}^2. \end{aligned}$$

Since $\mathbf{w}(0, \mathbf{x}) = \mathbf{0}$, from this inequality we deduce that there exists $T > 0$ such that

$$(2.23) \quad \|\mathbf{w}\| \leq c \max_{0 < t \leq T} \delta\tilde{p}.$$

Finally, the uniqueness of the solution can be proven by contradiction. Assume that there exist two solutions associated with the same data set. We can repeat steps (2.21), (2.22) with $\delta\tilde{p} = 0$. Then (2.23) implies that the solution is unique, and this concludes our proof. $\square$

Remark 2.1. Since $\mathbf{w}$ is a solenoidal function, we have the trace inequality (see [10] and (2.13))

$$\left| \int_{\Gamma_i} \mathbf{w} \cdot \mathbf{n} d\gamma \right| \leq c_{12} \|\mathbf{w}\|.$$

Together with (2.23) it implies, in particular, that the flow rates on the boundaries Γ_i are continuous functions of the pressure data.

3. Lumped parameters models. Modeling the whole circulatory system in terms of the Navier–Stokes equations as done in the previous section is not affordable in practice due to its prohibitive computational cost. In this context, simplified models are often applied. They are less accurate, as they cannot catch local details; however,

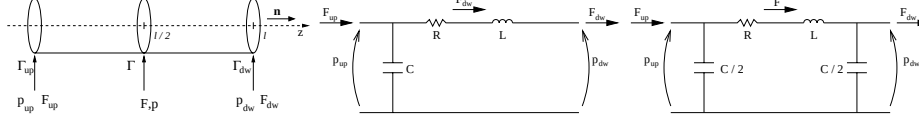


FIG. 3.1. Location of the variables of interest upstream, downstream, and in the middle of a cylindrical vessel featuring finite length l (left). A lumped $\mathcal{L}$ -network (center) and a π -network (right) are shown; the former network is equivalent to a compliant pipe with F_{up} and p_{dw} prescribed on the upstream and downstream sections, respectively; the latter network is equivalent to a compliant pipe with both the flow rates prescribed on both the upstream and downstream sections.

they can correctly account for systemic phenomena (like pressure pulse propagation or flow rate distribution at different environments) and have the fascination of being computationally very cheap.

The earliest and most simple global models considered in the literature are based on a representation of the circulatory system as a network of “compartments.” The mathematical description of the problem moves from an analogy with electric circuits. Actually, in the 1960s electric circuits were physically built as “circulation analog simulators” (see [27]). On the ground of this analogy, a systemic representation of the circulation is given by a set of ODEs. In the multiscale perspective of the present work, such equations need to be numerically linked to the Navier–Stokes model set on a specific local region (see, e.g., [16]).

In the following subsection we first introduce the lumped parameter model of a very simple compartment given by a compliant cylindrical pipe. This will help to clarify the physical meaning of the lumped parameters we are dealing with. Then we will give a short outline of the lumped description of districts of specific interest (such as the heart, the heart valves, etc.) that are assembled in a network of compartments. Finally, we will discuss the most relevant mathematical features of the associated system of ODEs in view of the subsequent analysis of the multiscale model.

3.1. Lumped models for a cylindrical vessel. Let us consider a cylindrical compliant pipe of length l and radius (at rest) R_0 . Denote by p_{up} and p_{dw} the mean pressure on the upstream and downstream sections, respectively. Moreover, we set

$$F_{up}(t) = \int_{\Gamma_{up}} \mathbf{u} \cdot \mathbf{n} d\gamma, \quad F_{dw}(t) = \int_{\Gamma_{dw}} \mathbf{u} \cdot \mathbf{n} d\gamma$$

as the upstream and downstream flow rates, respectively. We denote by $F(t)$ and $\bar{p}(t)$ the flow rate and the mean pressure at the midpoint of the pipe, respectively (Figure 3.1 left).

Let us integrate the Navier–Stokes equations (2.2) over the cylindrical domain Ω . In particular, we consider the axial component of the velocity, which we assume to be $\mathbf{u}_z$, and the corresponding momentum equation. Let us neglect the nonlinear convective term and the circumferential and axial components of the wall displacement. Moreover, following Westerhoff et al. (see [27]), we assume that

$$\int_{\Omega} -\nu \Delta \mathbf{u}_z d\omega \approx RF$$

with

$$(3.1) \quad R = \frac{8\nu l}{\pi R_0^4}.$$

This approximation is justified since it holds exactly in the case of the Hagen–Poiseuille flow for cylindrical pipes (see, e.g., [14]). Moreover, we assume that the pressure $\bar{p}$ is proportional to the radius of the pipe in such a way that the time derivative of $\bar{p}$ is proportional to the mean boundary radial velocity $\bar{\mathbf{u}}_{radial}$. Following [19], we set, in particular,

$$\frac{d\bar{p}}{dt} = \frac{Eh}{R_0^2(1-\xi^2)} \bar{\mathbf{u}}_{radial},$$

where ξ , E , and h are the Poisson ratio, the Young modulus, and the wall thickness. On the other hand, the integration of the mass conservation equation over the domain Ω yields

$$F_{up}(t) - F_{dw}(t) = \int_{\Gamma_{wall}} \mathbf{u}_{radial} d\gamma \approx 2\pi R_0 l \bar{\mathbf{u}}_{radial}.$$

Finally, assume that F_{up} and p_{dw} are prescribed, and set $p_{up} = \bar{p}$ and $F_{dw} = F$, which is reasonable for short pipes. With these assumptions, the integration of the Navier–Stokes equations over the domain Ω leads to the system

$$(3.2) \quad \frac{d\mathbf{y}}{dt} = A\mathbf{y} + \mathbf{b}(F_{up}, p_{dw})$$

being $\mathbf{y} = (p_{up}, F_{dw})^T$,

$$A = \begin{pmatrix} 0 & -\frac{1}{C} \\ \frac{1}{L} & -\frac{R}{L} \end{pmatrix}, \quad \mathbf{b}(F_{up}, p_{dw}) = \begin{pmatrix} \frac{1}{C} F_{up} \\ -\frac{1}{L} p_{dw} \end{pmatrix},$$

where

$$(3.3) \quad L = \frac{\rho l}{\pi R_0^2}, \quad C = \frac{3\pi R_0^3(1-\xi^2)l}{2Eh}.$$

This system can be represented as an electric circuit by the $\mathcal{L}$ -network shown in Figure 3.1 (center).

Now, suppose that $F_{up}(t)$ and $F_{dw}(t)$ are prescribed. This case can be faced by splitting the whole pipe into the sequence of two pipes (whose length is $l/2$) and proceeding in a way similar to the way we have followed above. Under the same simplifying assumptions we resort to the system (see, e.g., [16, 27])

$$(3.4) \quad \frac{d\mathbf{y}}{dt} = A\mathbf{y} + \mathbf{b}(F_{up}, F_{dw}), \quad t > 0,$$

where $\mathbf{y} = (p_{up}, p_{dw}, F)^T$ is the “state vector” (which depends on t),

$$A = \begin{pmatrix} 0 & 0 & -2/C \\ 0 & 0 & 2/C \\ 1/L & -1/L & -R/L \end{pmatrix}, \quad \mathbf{b}(F_{up}, F_{dw}) = \begin{pmatrix} 2F_{up}/C \\ -2F_{dw}/C \\ 0 \end{pmatrix}.$$

More precisely, the first two equations are obtained by the mass conservation equation integrated over the two pipes with length $l/2$. The third equation comes from the

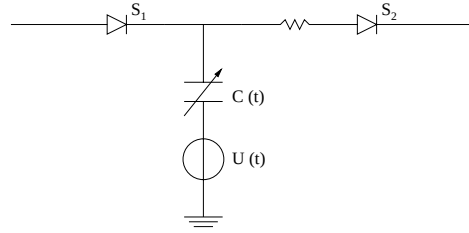


FIG. 3.2. Scheme of a ventricle, represented by a pressure source $U(t)$ and a compliance $C(t)$ changing in time. The two ventricular valves are modeled by two diodes, S_1 and S_2 .

integration of the momentum conservation equation. Equations (3.4) can be equivalently regarded as the mathematical description of an electric circuit, which is known as a π -network (see Figure 3.1 right).

Note that, in the hydraulic/electric analogy, pressure and flow rate correspond to the electric voltage and current, respectively; moreover, the resistor R is related to blood viscosity, the inductor L to the inertia of blood, and the capacitor C to the wall compliance.

3.2. Network models for the circulatory system. A network model of the circulatory system can be regarded as the assembly of many elementary bricks, such as the one in Figure 3.1. However, a specific description is required for the heart action of providing the energy necessary for blood for circulation in the whole system. The heart can be considered as a couple of pumps. In particular, in [21], each ventricle is represented as a compliant vessel whose compliance $C(t)$ changes in time. Moreover, in order to describe the heart action, in the electric/hydraulic analogy every valve behavior is modeled by a *diode* for the current, driven by the value of the applied voltage drop. Precisely, one ventricle is represented by a simple brick as in Figure 3.2, where two diodes, S_1 and S_2 , and a time dependent capacitance $C(t)$ are represented (see also [15]).

Altogether, the whole circulation can be described by a differential system whose abstract form reads

$$(3.5) \quad \begin{cases} \frac{d\mathbf{y}}{dt} = A(\mathbf{y}, t)\mathbf{y} + \mathbf{r}(\mathbf{y}, t), & t > 0, \\ \mathbf{y}(0) = \mathbf{y}_0, \end{cases}$$

where $\mathbf{y} \in \mathbb{R}^m$ is the vector of state variables, and $A(\mathbf{y}, t) \in \mathbb{R}^{m \times m}$ and $\mathbf{r}(\mathbf{y}, t) \in \mathbb{R}^m$ are a matrix and a vector whose dependence on $\mathbf{y}$ is due to the presence of diodes, while the dependence on time has to be attributed to the heart ventricle's compliances and the source term $\mathbf{U}(t)$.

Specific instances of systems like the one in (3.5) can be found, e.g., in [11, 13]. In the quoted references, lumped parameters systems featuring, respectively, 16 and 48 state variables for the description of the whole circulation have been proposed for a specific investigation of the coronary bed.

4. The geometrical multiscale model. We wish now to represent the whole system by an electric circuit except on a specific region Ω , where blood flow is modeled by the Navier–Stokes equations.

For the sake of simplicity, the compliance of the vessel wall is neglected; hence Ω is constant in time. The mean pressure and the flow rate through each artificial

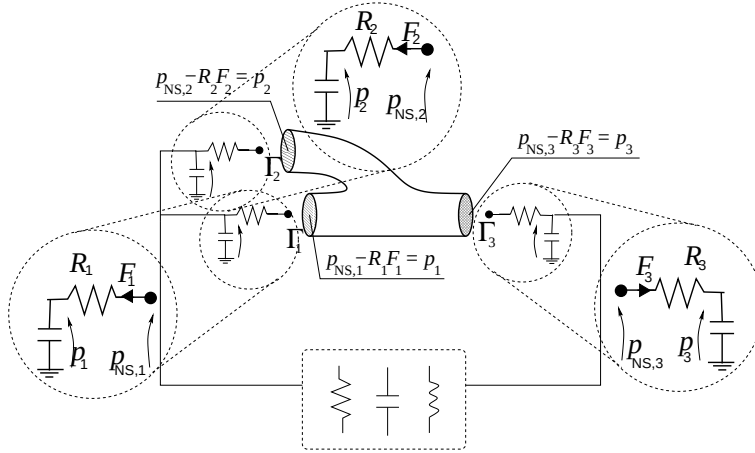


FIG. 4.1. Scheme of coupling between the whole system and a local district. The lumped representation of the three bridging regions at the interfaces with the Navier–Stokes model is highlighted in the dashed circles. This representation is related to the interface conditions (4.4).

section Γ_i of Ω ($i = 1, \dots, \bar{n}$) are denoted by p_i and F_i . We will set $\mathbf{F} = (F_i)_{i=1, \dots, \bar{n}}$, where $F_i(t) = \int_{\Gamma_i} \mathbf{u} \cdot \mathbf{n} d\gamma$.

Let us assume that the network faces the district Ω by capacitors C_i and resistors R_i ($i = 1, \dots, \bar{n}$) as shown in Figure 4.1. In particular, we have put in evidence the representation in terms of a network of the vascular regions in the immediate neighborhood of the 3D model. We denote these parts of the lumped network by *bridging regions*. In this picture, we have three bridging regions corresponding to the three inflow/outflow of Ω . The Navier–Stokes equations and the lumped network are actually coupled by means of interface conditions involving the flow rates and the mean pressure values in the bridging regions.

More precisely, we denote again by $\mathbf{y} \in \mathbb{R}^m$ the state vector of the circuit at hand, and, for the sake of notation, we will suppose that the first $\bar{n}$ state variables y_i ($i = 1, \dots, \bar{n}$, $\bar{n} < m$) correspond to the interface pressures $p_i(t)$. Due to the specific lumped representation of the bridging regions, the mean pressure on the Navier–Stokes interfaces is given by

$$(4.1) \quad p_i(t) - R_i F_i(t) \quad (i = 1, \dots, \bar{n}).$$

In fact, the term $-R_i F_i(t)$ plays the role of a correction of the pressure p_i , which is the external normal force applied in the usual Neumann condition. As we have pointed out, on the lumped model variable p_i has the meaning of an average pressure over the bridging region. If we assume p_i to be representative of the value of the pressure at the interface, then R_i can be set equal to 0 and the condition at hand reduces to consider the mean pressure problem addressed in [10]. However, a more accurate guess for the pressure at the interface can be enforced by introducing a correcting term. The latter is a pressure drop that we have taken proportional to the flow rate, as we have in a Hagen–Poiseuille flow (which is one of the main assumptions behind the derivation of our zero-dimensional model), yielding (4.1).

We will assume that the network is modeled by the differential system (3.5). In particular, we set

$$\mathbf{r}(\mathbf{y}, t) = \mathbf{r}_H(\mathbf{y}, t) + \mathbf{b}(\mathbf{F}(t)),$$

where $\mathbf{r}_H$ accounts for the heart action, $\mathbf{b}$ involves the variables at the external terminals in such a way that

$$(4.2) \quad \begin{cases} b_i = C_i F_i & \text{for } i = 1, \dots, \bar{n}, \\ b_k = 0 & \text{otherwise,} \end{cases}$$

and C_i are suitable constants depending on the capacitors at the interfaces.

The coupled problem reads therefore as follows.

PROBLEM 4.1. *Given $\mathbf{u}_0 \in \mathbf{V}^*$ and $\mathbf{y}_0 \in \mathbb{R}^m$, find $\mathbf{u} \in L^2(0, T; \mathbf{V})$ and $\mathbf{y} \in \mathbf{L}^\infty(0, T)$ such that for every $t > 0$*

$$(4.3) \quad \begin{cases} \frac{d\mathbf{y}}{dt} = A(\mathbf{y}, t)\mathbf{y} + \mathbf{r}_H(\mathbf{y}, t) + \mathbf{b}(\mathbf{F}(t)), \\ \left(\frac{\partial \mathbf{u}}{\partial t}, \varphi \right) + a(\mathbf{u}, \varphi) + b(\mathbf{u}, \mathbf{u}, \varphi) = c(\varphi) \quad \forall \varphi \in \mathbf{V}^* \end{cases}$$

with $\mathbf{y}(0) = \mathbf{y}_0$, $\mathbf{u}|_{t=0} = \mathbf{u}_0$, where, as stated above, the first $\bar{n}$ entries of the state vector $\mathbf{y}$ correspond to the interface pressures in such a way that the mean pressures at the interface boundaries Γ_i are given by

$$(4.4) \quad p_{NS,i}(t) = p_i(t) - R_i F_i(t) = y_i(t) - R_i F_i(t), \quad i = 1, \dots, \bar{n},$$

and, by definition, the entries of the vector $\mathbf{F}$ are given by

$$(4.5) \quad F_i(t) = \int_{\Gamma_i} \mathbf{u} \cdot \mathbf{n} d\gamma, \quad i = 1, \dots, \bar{n}.$$

In (4.3), the bilinear form $a(\cdot, \cdot)$, the trilinear form $b(\cdot, \cdot, \cdot)$, as well as the functional $c(\cdot)$, are the same as introduced in (2.7a), (2.7b), (2.7c), (2.7d). Observe that $c(\cdot)$ in (4.3) is a function of the vector $\mathbf{y}$ and, in particular, of its first $\bar{n}$ components.

In Problem 4.1 we match a local 3D model, for a domain of the order of a few centimeters, with a systemic model, covering a much wider region. The level of detail provided by the two models is obviously not the same. This difference is evident and relevant at the interfaces. On one hand, the ODE model considers mean data, namely $p_i(t)$ and $F_i(t)$, as specified in (4.4) and (4.5). On the other hand, the Navier–Stokes model would require pointwise conditions like (2.6). In Problem 4.1, this difference is in fact faced by a natural treatment of the boundary conditions, as suggested, in a similar context, in [10]. In particular, the variational formulation for the Navier–Stokes problem considered in (4.3) accounts for the interface data (4.4) in a natural way, which corresponds to force a pointwise relation like (2.6), provided that $K_i = R_i A_i$, A_i being the section area. Besides, (4.5) states the relation between the mean data of velocity considered by the systemic side and the pointwise velocity field of the Navier–Stokes equations. In the next section, we will see that this strategy ensures the well posedness of the coupled problem. In section 5.3, however, we will return to this issue and, in particular, to the impact of substituting (4.4) with (2.6).

5. Well-posedness analysis of the multiscale model. In this section, we prove a local-in-time existence result for Problem 4.1 by reformulating it as a fixed-point problem. We will prove that the associated fixed-point map $\mathcal{T}$ is a *compact operator* from a bounded subset of a Banach space into itself so that the classical Schauder theorem can be applied.

It is worthwhile to observe a similar fixed-point approach that we have adopted in [16] for the numerical solution of this multiscale problem.

5.1. An auxiliary linear lumped model. In order to define the operator $\mathcal{T}$, we introduce the following ordinary differential system, which is a linear counterpart of (4.3a):

$$(5.1) \quad \begin{cases} \frac{d\mathbf{y}}{dt} = A(\mathbf{z}, t)\mathbf{y} + \mathbf{r}_H(\mathbf{z}, t) + \mathbf{b}(\mathbf{F}(t)), & t > 0, \\ \mathbf{y}(0) = \mathbf{y}_0, \end{cases}$$

where the vector of fluxes $\mathbf{F}(t)$, matrix $A(\mathbf{z}, t)$, and vectors $\mathbf{r}_H(\mathbf{z}, t)$ and $\mathbf{b}(\mathbf{F}(t))$ are defined as in (4.3) and $\mathbf{z} \in \mathbf{L}^\infty(0, T)$ is a given vector function.

Suppose that there exist two constants, $C_A, C_H > 0$, such that

$$(5.2) \quad \|A(\mathbf{z}, t)\|_2 \leq C_A \quad \text{and} \quad |\mathbf{r}_H(\mathbf{z}, t)| \leq C_H \quad \forall \mathbf{z} \in \mathbf{L}^\infty(0, T), \forall t \in (0, T).$$

Finally, define $c_0 = \max_i(C_i^2)$, where C_i are given in (4.2), so that

$$(5.3) \quad |\mathbf{b}(\mathbf{F}(t))|^2 \leq c_0 \sum_{i=1}^{\bar{n}} |F_i(t)|^2 \quad \forall t > 0.$$

Classical results (see, e.g., [20]) yield the existence and uniqueness of the solution of the linear system (5.1). Moreover, using the Gronwall lemma, we obtain

$$(5.4) \quad |\mathbf{y}(t)|^2 \leq \left(|\mathbf{y}_0|^2 + C_H^2 t + c_0 \sum_{i=1}^{\bar{n}} \int_0^t |F_i(s)|^2 ds \right) \exp(2(1 + C_A)t) \quad \forall t > 0.$$

5.2. The main result. For any $\sigma > 0$ we consider the subset of $\mathbf{L}^\infty(0, T)$:

$$(5.5) \quad \mathcal{B}_{\sigma, T} = \{\mathbf{z} \in \mathbf{L}^\infty(0, T) : |\mathbf{z}| \leq \sigma\}.$$

Consider Problem 2.1 and a vector $\mathbf{z} \in \mathcal{B}_{\sigma, T}$. Denote by $\mathcal{L}_{NS}$ the operator mapping $\mathbf{z}(t)$ into the solution $\mathbf{u}(\mathbf{x}, t)$ of Problem 2.1:

$$\mathbf{u} = \mathcal{L}_{NS}\mathbf{z}.$$

More precisely, we solve Problem 2.1 with the pressure data provided by the first $\bar{n}$ components of $\mathbf{z}$. Then we set $\mathbf{F}(t)$ the vector whose entries $F_i(t)$ are functions of $\mathbf{u}(\mathbf{x}, t)$ according to (4.5) and solve the linear ordinary differential system (5.1). In this way, we implicitly define the operator $\mathcal{L}_{NET}$, mapping $\mathbf{u}(\mathbf{x}, t)$ into $\mathbf{y}(t)$,

$$\mathbf{y} = \mathcal{L}_{NET}\mathbf{u},$$

and therefore a composite operator,

$$\mathcal{T} = \mathcal{L}_{NET} \cdot \mathcal{L}_{NS} : \mathcal{B}_{\sigma, T} \rightarrow \mathbf{L}^\infty(0, T), \quad \mathbf{z} \rightarrow \mathbf{y} = \mathcal{T}\mathbf{z}.$$

At this stage, we can easily see that if $\mathbf{y}$ is a fixed point of $\mathcal{T}$, i.e., $\mathbf{y} = \mathcal{T}\mathbf{y}$, then the couple $(\mathbf{y}, \mathbf{u} = \mathcal{L}_{NS}\mathbf{y})$ is a solution of Problem 4.1. We are therefore left to prove that $\mathcal{T}$ has a fixed point.

PROPOSITION 5.1. *Suppose that the initial data $\mathbf{y}_0$ and $\nabla \mathbf{u}_0$ are small enough and, in particular, that*

$$(5.6) \quad |\mathbf{y}_0| < \sigma, \quad \|\nabla \mathbf{u}_0\| \leq \sigma_1,$$

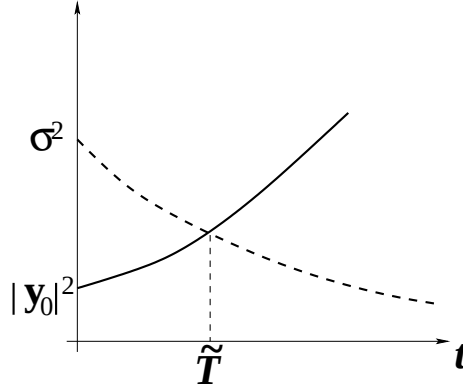


FIG. 5.1. Intersection between the curves $|y_0|^2 + C_H^2 t + \bar{n} c_0 c_5^2 \int_0^t \|\nabla \mathbf{u}(s)\|^2 ds$ (solid line) and $\sigma^2 \exp(-(2 + C_A)t)$ (dashed line).

where σ and σ_1 are chosen in such a way that Problem 2.2 is well posed as stated in Proposition 2.1. Moreover, suppose that (5.2) and (2.16) hold. Consequently, there exists $\tilde{T}$ ($0 < \tilde{T} \leq T$) such that $\mathcal{T}$ is a compact operator mapping $\mathcal{B}_{\sigma, \tilde{T}}$ into itself. Then $\mathcal{T}$ admits at least one fixed point in $\mathcal{B}_{\sigma, \tilde{T}}$.

Proof. Let $\mathbf{z}$ be a vector in $\mathcal{B}_{\sigma, T}$, and define $\mathbf{y} = \mathcal{T}\mathbf{z}$. Note that from (5.4) $\mathbf{y} \in \mathbf{L}^\infty(0, T)$. Let us prove that there exists a time interval $\mathcal{I} \equiv (0, \tilde{T})$ such that $\mathbf{y} \in \mathcal{B}_{\sigma, \tilde{T}}$ for $t \in \mathcal{I}$. In fact, this proves that $\mathcal{T}$ maps $\mathcal{B}_{\sigma, \tilde{T}}$ into itself. According to the notation of section 2, set $\tilde{p} = \sup_{t \geq 0} \sum_{i=1}^{\bar{n}} |p_i(t)|$, where, as usual, p_i are the first $\bar{n}$ entries of $\mathbf{z}$. Since $\sum_{i=1}^{\bar{n}} |p_i(t)|^2 \leq |\mathbf{z}(t)|^2$ for every t , it follows that $\tilde{p} \leq \sigma$. Then, according to Proposition 2.1, there exists a unique solution $\mathbf{u}$ of Problem 2.2 in a time interval $(0, T)$ such that

$$(5.7) \quad \|\nabla \mathbf{u}(t)\| \leq \sigma_1 \quad \forall t \in (0, T).$$

Furthermore, consider the system (3.5) with $F_i(t)$ as in (4.5). If φ is any function of $\mathbf{V}^*$ and Γ any subset of $\partial\Omega$, the inequality

$$(5.8) \quad \left| \int_{\Gamma} \varphi \cdot \mathbf{n} d\gamma \right| \leq c_5 \|\nabla \varphi\|$$

follows from the trace and the Poincaré inequalities. Then, owing to (5.7) and (5.8), $\mathbf{b}(\mathbf{F}(t))$ in (5.3) belongs to $\mathbf{L}^2(0, T)$, whence the Cauchy problem (5.1) is well posed in $(0, T)$. Denote by $\mathbf{y}$ its unique solution. In particular, since $|F_i(t)| \leq c_5 \|\nabla \mathbf{u}(t)\|$, from (5.4) it follows that

$$(5.9) \quad |\mathbf{y}(t)|^2 \leq \left(|\mathbf{y}_0|^2 + C_H^2 t + \bar{n} c_0 c_5^2 \int_0^t \|\nabla \mathbf{u}(s)\|^2 ds \right) \exp((2 + C_A)t).$$

Under assumption (5.6) it is possible to verify (see Figure 5.1) that there exists a positive $\tilde{T}$ such that the inequality

$$(5.10) \quad |\mathbf{y}_0|^2 + C_H^2 t + \bar{n} c_0 c_5^2 \int_0^t \|\nabla \mathbf{u}(s)\|^2 ds \leq \sigma^2 \exp(-(2 + C_A)t)$$

holds for all $0 \leq t \leq \tilde{T}$. Indeed, $\tilde{T}$ corresponds to the intersection point between the monotonically increasing curve $|\mathbf{y}_0|^2 + C_H^2 t + \bar{n} c_0 c_5^2 \int_0^t \|\nabla \mathbf{u}(s)\|^2 ds$ and the monotonically decreasing curve $\sigma^2 \exp(-(2 + C_A)t)$ (see Figure 5.1).

Thus, thanks to (5.9), we have

$$(5.11) \quad |\mathbf{y}(t)| \leq \sigma$$

for each $t \in (0, \tilde{T})$. Therefore, in $(0, \tilde{T})$ $\mathcal{T}$ maps $\mathcal{B}_{\sigma, \tilde{T}}$ into itself.

Moreover, $\mathcal{T} = \mathcal{L}_{NET} \cdot \mathcal{L}_{NS}$ is continuous. Indeed, this follows from the continuity of the two maps, $\mathcal{L}_{NS}$ and $\mathcal{L}_{NET}$; $\mathcal{L}_{NS}$ is continuous as proven in (2.23), while the continuity of $\mathcal{L}_{NET}$ is an immediate consequence of the continuity of $\mathbf{b}(\cdot)$ defined in (4.2) and the fact that the ordinary differential system at hand (5.1) is linear.

Having proven the continuity of $\mathcal{T}$, we now need to prove its compactness. Since the set $\mathcal{B}_{\sigma, \tilde{T}}$ is bounded by construction, we need simply to prove the compactness of the range of $\mathcal{B}_{\sigma, \tilde{T}}$ according to $\mathcal{T}$. We have already verified that this set is bounded. Moreover, from (5.1) we have for all $t \in \mathcal{I}$

$$\left| \frac{\partial \mathbf{y}}{\partial t} \right| = |A(\mathbf{z}, t)\mathbf{y} + r_H(\mathbf{z}, t) + \mathbf{b}(\mathbf{F})| \leq C_A \sigma + C_H + C \sigma_1 = M,$$

where the constant M does not depend on t . We can therefore conclude that the image of a bounded set in $\mathbf{L}^\infty(0, \tilde{T})$ according to the map $\mathcal{T}$ is bounded and equicontinuous. Thanks to the Ascoli–Arzelà theorem (see, e.g., [31]), this means that the image of a bounded set according to $\mathcal{T}$ is compact. Since $\mathcal{T}$ is continuous, we conclude that $\mathcal{T}$ is compact.

The (local-in-time) existence of the solution of the multiscale Problem 4.1 follows from the Schauder fixed-point principle ([31]). $\square$

It is worthwhile to point out that, if $(\mathbf{u}, \mathbf{y})$ satisfies Problem 4.1, then there exists a unique function $p \in L^2(0, \tilde{T}; L^2(\Omega))$ such that the couple $(\mathbf{u}, p)$ is the solution of the Navier–Stokes submodel. Indeed, the map $\mathcal{L}_p : \mathbf{V} \rightarrow \mathbb{R}$ such that $\mathcal{L}_p(\boldsymbol{\varphi}) = \frac{d}{dt}(\mathbf{u}, \boldsymbol{\varphi}) + a(\mathbf{u}, \boldsymbol{\varphi}) + b(\mathbf{u}, \mathbf{u}, \boldsymbol{\varphi})$ belongs to $\mathbf{V}'$ and vanishes on $\mathbf{V}^*$. Therefore, by well-known results (see, e.g., [8]) there exists a function $p \in L^2(\Omega)$ (which is unique up to an additive constant) such that $\mathcal{L}_p(\boldsymbol{\varphi}) = (p, \nabla \boldsymbol{\varphi})$ for all $\boldsymbol{\varphi} \in \mathbf{V}$. Thanks to conditions (2.6), p is the unique function such that $(\mathbf{u}, p)$ solves the Navier–Stokes problem.

Remark 5.1. The initial bounds assumed by hypothesis in (5.6) actually hold in $\mathcal{I}$, in the sense that

$$(5.12) \quad |\mathbf{y}(t)| \leq \sigma \quad \text{and} \quad \|\nabla \mathbf{u}(t)\| \leq \sigma_1$$

for each $t \in (0, \tilde{T})$. This is an immediate consequence of the estimates found in proving Propositions 2.1 and 5.1.

Remark 5.2. The introduction of the auxiliary linear model (5.1) is not strictly necessary to prove our main result. Indeed, we could have considered instead the original coupled model (4.3), which includes the nonlinear lumped model. However, the use of the auxiliary linear model has the advantage of allowing a much simpler approach without assuming severe restrictions on the coefficients of the matrix A , which would be necessary when analyzing the nonlinear lumped model. Obviously, the fixed point of the models (the linearized one and the original one) are the same.

Remark 5.3. In section 3 we have pointed out that systemic models can be obtained by considering the circulation as a set of regions connected to one another. Starting from a 3D description of blood flow in each region (based on the Navier–Stokes equations), we obtain a simplified mathematical description carrying out a

space average of the basic equations of mass and momentum conservation. The actual values of the parameters depend on the geometrical and physical features of the compartment at hand (see (3.3)). As an example, in the case of a compliant straight pipe, the value of the resistance parameter R is a function of the blood viscosity ν , the radius of the pipe R_0 , and the length of the compartment l .

In the multiscale model, interface conditions (4.4) involve the parameters of the lumped network. In particular, conditions (4.4) depend on the resistance parameters R_i featuring the bridging regions interposed between the local model and the remainder of the circulation. In this context, assumption (2.16) can be regarded as a constraint on the geometry of the bridging regions. In particular, suppose that these regions are cylindrical vessels. Owing to (3.3), (2.16) reads $16lc_2c_4 < \pi R_0^4$. In fact, the assumption (2.16) can be reinterpreted as a constraint on the length l of the bridging region which must be sufficiently small. In other words, a suitable lumped representation, where the bridging districts have been suitably chosen, can always satisfy (2.16).

5.3. Different splittings of the coupled problem. In the proof of Proposition 5.1 we have analyzed the coupled problem by means of a splitting of the multiscale problem into a local Navier–Stokes problem and a global ordinary differential system. In this context, at the first step of the splitting, the interface conditions (4.4) have played the role of boundary conditions for the Navier–Stokes problem. On the other hand, the interface conditions (4.5) have been accounted for in the second step through the forcing terms of the lumped parameters system.

As pointed out in section 4, there is a difference between the data accounted for at the interfaces of the two models. We are now in a position of specifying how this difference is faced. Indeed, (4.4) is not enough as boundary condition for a 3D Navier–Stokes problem, since it prescribes average data on the interfaces, while three conditions would be required pointwise [25]. However, the selected variational formulation introduced in Problems 2.1 and 4.1 corresponds to the natural treatment of *defective* conditions first introduced for the mean pressure drop problem in [10]. More precisely, the weak formulation (4.3) embodies condition (2.6) as a natural condition. In the special case of a cylindrical domain with boundary sections orthogonal to the vessel axis, condition (2.6) does actually reduce to (4.4), as the viscous forces $\nu \nabla \mathbf{u} \cdot \mathbf{n}$ vanish on Γ_i . Although the latter circumstance does not occur in domains of general shape, nonetheless the extra conditions stemming from (2.6) with respect to (4.4) are satisfied in a weak sense and therefore yield mild perturbation on the associated solution.

The role of the interface conditions in the splitting procedure is in fact naturally driven by the specific topology of the network at the interfaces. In our case, the interface flow rates are not state variables of the lumped system, and, therefore, they are well suited to play the role of a forcing term for the ordinary differential system. However, depending on the choice of the bridging regions, the matching between the network and the Navier–Stokes system could be pursued, for instance, by interchanging the role of flux and pressure at the interfaces.

When splitting the coupled problem, we should suppose that the flow rates are provided to the Navier–Stokes system by the network, which in turn receives pressure data. For instance, in the network configuration of Figure 5.2, the interface pressure is not a state variable of the lumped system, so it is a good candidate for being a forcing term of the ordinary differential system provided by the Navier–Stokes solution. On the other hand, the interface flow rates, which in the electric analogy correspond to the

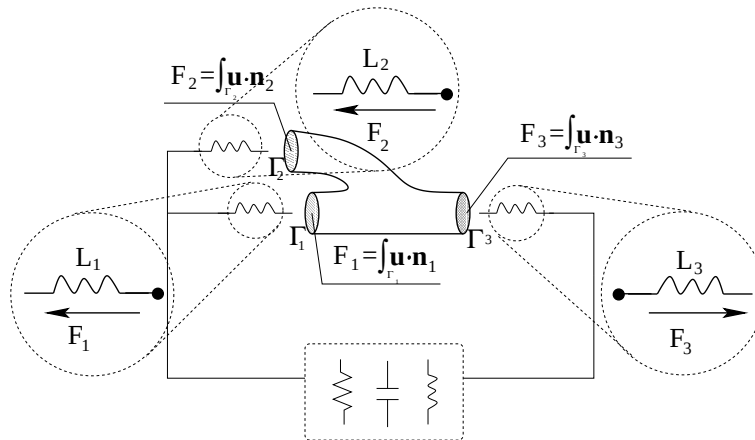


FIG. 5.2. Scheme of coupling between the whole system and a local district where the bridging regions are given by inductors.

current at the interfaces and are state variables for the system, become boundary data for the Navier–Stokes problem. In a way similar to the (generalized) mean pressure problem (conditions (4.4)), the prescription of boundary fluxes is not enough to make the Navier–Stokes problem well posed. The defective data still need to be suitably completed. A theoretical analysis of this problem has been carried out in [10], while a reformulation of this problem that is more suitable for the numerical approximations is provided in [6]. The results in Proposition 5.1 can be extended to this case as well. Actually, a fixed-point approach can be adopted in the proof, in a completely similar way to the one adopted here, starting from the well-posedness analysis of the individual submodels.

Remark 5.4. Non-Newtonian rheology. In the setup of the local model we have assumed that blood behaves like a Newtonian fluid. It could be of some interest to address to what extent our analysis should be modified if we account for a more realistic rheology. Blood is actually a viscoelastic shear thinning fluid, and this behavior can be described by different models, with different levels of accuracy. One of the most sophisticated models, in this respect, is the modified Oldroyd-B model proposed by Yelaswarapu (see [29]). Accounting for a non-Newtonian rheology would translate to a modification of the local Navier–Stokes model. This change would affect the matching conditions between the 3D and the lumped models. Precisely, the condition of equilibrium between forces would still be valid; however, the term on the left-hand side of (2.5) should take the more general form for Σ .

At the level of the analysis, prior to considering the multiscale coupling, we should focus on the well posedness of the local non-Newtonian models. So far, very little has been done concerning the mathematical theory of non-Newtonian models relevant for blood flow. Some results in this respect have been proven in [22, 23, 24]. However, results that could be applied to the case of our local model are not available at the moment, at least whenever the viscoelasticity of blood is considered, which is particularly relevant, for instance, in capillaries. Besides, the properties of $\mathcal{L}_{NSnonNewt}$ that would make applicable the Schauder fixed-point theorem should be investigated. These tasks are not simple to accomplish, and we think that they deserve by themselves specific investigations in future works.

Remark 5.5. Compliant vessels. If we remove the hypothesis of rigid vessels (2.4),

the multiscale model should be modified accordingly. In particular, besides (4.4) and (4.5), we should add a further condition which provides the 3D model information on the vessel wall displacement. Such a condition is actually a by-product of the constitutive law which is hidden behind the derivation of the lumped model and, in particular, in the specific expression of constant C in (3.3). The analysis of the (local) fluid structure interaction problem in three dimensions is still an open and challenging problem. To our knowledge, there are no results available concerning compliant models for vascular flow addressing realistic conditions (e.g., nonperiodic boundary conditions) in three dimensions. It is therefore obvious that our analysis cannot be extended since a basic building block (local analysis) is missing. We think that also the extension to compliant models is not trivial at all and should deserve a specific investigation.

6. A numerical test case. In this section, we illustrate some numerical results on a test case based on the Womersley solution for the Navier–Stokes equations (see [28]).

Consider a flow between two infinite planes where a periodic-in-time pressure gradient is prescribed. Let the two planes be at distance d in the frame of reference (x_1, x_2) . In particular, if the pressure gradient is given by

$$\nabla p = A \sin(\omega t),$$

the flow field can be analytically determined, providing in 2D the analogue of the Womersley solution that has been obtained for a 3D cylindrical domain (see [28, 17]). In particular, the transversal velocity u_2 is null, while the axial velocity u_1 is

$$(6.1) \quad u_1 = \sum_{k=0}^{\infty} \gamma_{2k+1} \sin\left(\frac{(2k+1)\pi}{d} x_2\right),$$

where, setting $\beta = \frac{\nu\pi^2}{d^2}$, $\gamma_i = \frac{4A}{\pi i(i^4\beta^2 + \omega^2)}(i^2\beta \sin(\omega t) + \omega e^{-i^2\beta t} - \omega \cos(\omega t))$. Observe that (like for the Poiseuille flow) the nonlinear term in the Navier–Stokes equations vanishes, independently of A and ω , since, due to the particular morphology and the boundary data independent of x_2 , the velocity field is orthogonal to its gradient. Therefore, for more complex periodic pressure differences, the analytical solution can be determined by expanding the pressure drop in a Fourier series and superimposing the contribution of each term accordingly.

The variational strategy illustrated above has been successfully adopted in [26] for computing the Womersley solution (6.1) on a couple of finite length of planes, i.e., a 2D rectangular pipe, featuring an upstream and a downstream section. In the multiscale context, we have coupled such a 2D rectangular pipe with a lumped parameter model, given in Figure 6.1. The pressure difference applied to the 2D region is periodic (see Figure 6.2), so we can compute analytically the local flow Womersley solution. In Table 6.1 we illustrate the $L^2(\Omega)$ errors at different instants of the heart beat.

These numerical results confirm the correctness of our approach and the well posedness of the multiscale model stated in the present work.

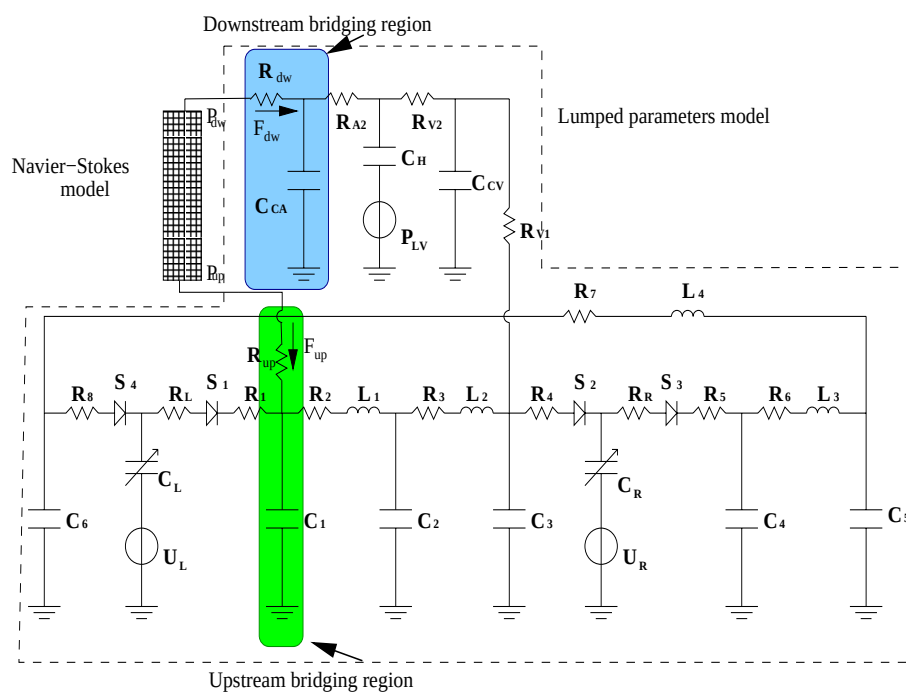


FIG. 6.1. Representation of the multiscale test case.

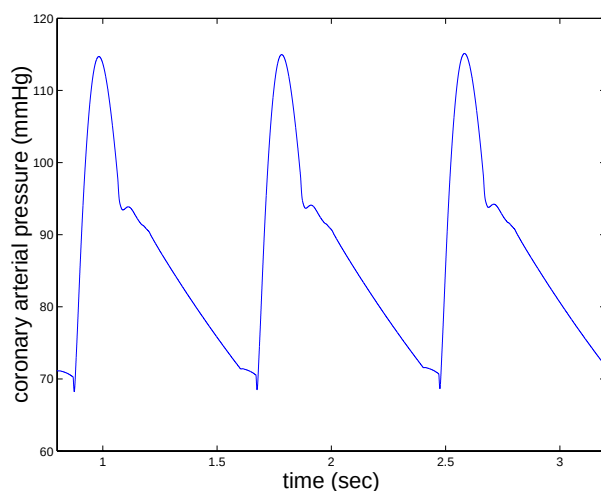


FIG. 6.2. Pressure drop of the multiscale simulation applied to the two-dimensional (2D) region.

Beyond this academic test case, numerical results obtained in more realistic contexts, still based on the approach of the present work, can be found in [16] and in [2, 3, 4]. In these references the adoption of geometrical multiscale models has revealed very promising for analyzing, by means of numerical simulations, the dynamics of flow patterns in morphologically complex vascular districts in the context of pediatric surgery. The goal was to identify some “optimal” design for surgical interventions. Figure 6.3 (taken from [4], courtesy of the authors) illustrates some



FIG. 6.3. *Simulation of complex vascular districts: velocity fields at different instants of the heart beat. These are the 3D results of multiscale simulations, allowing a realistic boundary data prescription. See [4] for more details.*

numerical results obtained with the geometrical multiscale approach of the present work for the ascending aorta region after a surgical intervention (the so-called MBT shunt, Modified Blalock–Taussig shunt). Thanks to the multiscale approach, it has

TABLE 6.1

Relative errors in the Womersley test case during a heart beat. The heart beat period is fixed to 0.8 s.

t	$\ \mathbf{u}_{\text{Womersley}} - \mathbf{u}_{\text{computed}}\ $
T_{beat}	$\ \mathbf{u}_{\text{Womersley}}\ $
0.125	6.0×10^{-4}
0.250	2.9×10^{-6}
0.375	2.4×10^{-5}
0.500	1.4×10^{-4}
0.625	1.1×10^{-6}
0.750	1.7×10^{-6}
0.875	1.8×10^{-6}
1.000	2.5×10^{-6}

been possible to compute physiological velocity profiles and secondary flow patterns which were not captured in previous simulations, based on the prescription of incorrect (even if realistic) essential boundary velocity data.

Acknowledgments. The authors wish to thank Prof. S. Salsa for several useful suggestions, Dr. S. Ragni for having collaborated on an early version of this paper and carried out the numerical simulations of Table 6.1, and Prof. G. Dubini and Dr. F. Migliavacca for having provided the numerical results of Figure 6.3.

REFERENCES

- [1] J. P. ARCHIE AND R. P. FELDTMAN, *Critical stenosis of the internal carotid artery*, Surgery, 89 (1981), pp. 67–70.
- [2] K. LAGANÀ, G. DUBINI, F. MIGLIAVACCA, R. PIETRABISSA, G. PENNATI, A. VENEZIANI, AND A. QUARTERONI, *Multiscale modelling as a tool to prescribe realistic boundary conditions for the study of surgical procedures*, Biorheology, 39 (2002), pp. 359–364.
- [3] G. DUBINI, F. MIGLIAVACCA, R. PIETRABISSA, A. QUARTERONI, S. RAGNI, AND A. VENEZIANI, *From the global cardiovascular system hemodynamics down to the local blood motion: Preliminary applications of a multiscale approach*, in Proceedings of European Congress on Computational Methods in Applied Sciences and Engineering, Barcelona, Spain, 2000.
- [4] G. DUBINI AND F. MIGLIAVACCA, *manuscript*.
- [5] L. FORMAGGIA, J. F. GERBEAU, F. NOBILE, AND A. QUARTERONI, *On the coupling of 3D and 1D Navier-Stokes equations for flow problems in compliant vessels*, Comput. Methods Appl. Mech. Engrg., 191 (2001), pp. 561–582.
- [6] L. FORMAGGIA, J. F. GERBEAU, F. NOBILE, AND A. QUARTERONI, *Numerical treatment of defective boundary conditions for the Navier-Stokes equations*, SIAM J. Numer. Anal., 40 (2002), pp. 376–401.
- [7] L. FORMAGGIA, F. NOBILE, A. QUARTERONI, AND A. VENEZIANI, *Multiscale modelling of the circulatory system: A preliminary analysis*, Comput. Vis. Sci., 2 (1999), pp. 75–83.
- [8] V. GIRAUT AND P. RAVIART, *Finite Element Methods for Navier-Stokes Equations. Theory and Algorithms*, Springer Ser. Comput. Math. 5, Springer-Verlag, New York, 1986.
- [9] J. G. HEYWOOD AND R. RANNACHER, *Finite element approximation of the nonstationary Navier-Stokes problem. I. Regularity of solutions and second-order error estimates for spatial discretization*, SIAM J. Numer. Anal., 19 (1982), pp. 275–311.
- [10] J. HEYWOOD, R. RANNACHER, AND S. TUREK, *Artificial boundaries and flux and pressure conditions for the incompressible Navier-Stokes equations*, Internat. J. Numer. Methods Fluids, 22 (1996), pp. 325–352.
- [11] F. INZOLI, F. MIGLIAVACCA, AND S. MANTERO, *Pulsatile flow in an aorto-coronary bypass 3-D model*, in Biofluid Mechanics: Proceedings of the 3rd International Symposium, D. Liepsch, ed., VDI-Verlag, Dusseldorf, Germany, 1994, pp. 455–463.
- [12] J. L. LIONS AND E. MAGENES, *Problèmes aux Limites non Homogènes et Applications*, 1, Dunod, Paris, 1969.
- [13] G. PENNATI, F. MIGLIAVACCA, G. DUBINI, R. PIETRABISSA, AND M. R. DE LEVAL, *A mathemat-*

- ical model of circulation in the presence of the bidirectional cavopulmonary anastomosis in children with a univentricular heart*, Med. Eng. Phys., 19 (1997), pp. 223–234.
- [14] F. WHITE, *Viscous Fluid Flow*, McGraw-Hill, New York, 1987.
 - [15] F. HOPPENSTEADT AND C. PESKIN, *Mathematics in Medicine and the Life Sciences* Springer-Verlag, New York, 1992.
 - [16] A. QUARTERONI, S. RAGNI, AND A. VENEZIANI, *Coupling between lumped and distributed models for blood flow problems*, Comput. Vis. Sci., 4 (2001), pp. 111–124.
 - [17] A. QUARTERONI, M. TUVERI, AND A. VENEZIANI, *Computational vascular fluid dynamics: problems, models and methods*, Comput. Vis. Sci., 2 (2000), pp. 163–197.
 - [18] A. QUARTERONI AND A. VALLI, *Domain Decomposition Methods for Partial Differential Equations*, Oxford University Press, Oxford, UK, 1999.
 - [19] V. C. RIDEOUT AND D. E. DICK, *Difference-differential equations for fluid flow in distensible tubes*, IEEE Trans. Bio-Medical Engrg., 14 (1967), pp. 171–177.
 - [20] G. BIRKHOFF AND G. C. ROTA, *Ordinary Differential Equations*, John Wiley and Sons, New York, 1989.
 - [21] K. SAGAWA, H. SUGA, AND K. NAKAYAMA, *Instantaneous pressure-volume ratio of the left ventricle versus instantaneous force-length relation of papillary muscle*, in Cardiovascular System Dynamics, J. Baan, A. Noordergraaf, and J. Raines, eds., MIT Press, Cambridge, MA, 1978, pp. 99–105.
 - [22] J. MALEK, K. R. RAJAGOPAL, AND M. RUZICKA, *Existence and regularity of solutions and the stability of the rest state for fluids with shear dependent viscosity*, Math. Models Methods Appl. Sci., 5 (1995), pp. 789–812.
 - [23] M. A. FONTELOS AND A. FRIEDMAN, *Stationary non-Newtonian fluid flows in channel-like and pipe-like domains*, Arch. Ration. Mech. Anal., 151 (2000), pp. 1–43.
 - [24] N. ARADA AND A. SEQUEIRA, *Strong Steady Solutions for Generalized Oldroyd-B Model with Shear-Dependent Viscosity in a Bounded Domain*, Preprint Department of Mathematics, Instituto Superior Tecnico, Lisboa (Portugal), 2002.
 - [25] R. TEMAM, *Navier–Stokes Equations and Nonlinear Functional Analysis*, CBMS-NSF Regional Conf. Ser. in Appl. Math. 66, SIAM, Philadelphia, 1995.
 - [26] A. VENEZIANI, *Boundary conditions for blood flow problems*, in Proceedings of the 2nd European Conference on Numerical Mathematics and Advanced Applications, Heidelberg, Germany, 1997, R. Rannacher et al., eds., World Scientific, River Edge, NJ, 1998, pp. 596–605.
 - [27] N. WESTERHOF, F. BOSMAN, C. J. DE VRIES, AND A. NOORDERGRAAF, *Analog studies of the human systemic arterial tree*, J. Biomech., 2 (1969), pp. 121–143.
 - [28] J. WOMERSLEY, *Method for the calculation of velocity, rate of flow and viscous drag in arteries when the pressure gradient is known*, J. Physiol., 127 (1955), pp. 553–563.
 - [29] K. YELASWARAPU, *Evaluation of Continuum Models for Characterizing the Constitutive Behavior of Blood*, Ph.D. thesis, University of Pittsburgh, Pittsburgh, PA, 1996.
 - [30] W. XIE, *Integral representations and L^∞ bounds for solutions of the Helmholtz equation on arbitrary open sets in $\mathbb{R}^2$ and $\mathbb{R}^3$* , Differential Integral Equations, 8 (1995), pp. 689–698.
 - [31] E. ZEIDLER, *Applied Functional Analysis*, Springer-Verlag, New York, 1991.